

A comparative assessment of machine learning algorithms in the IoT-based network intrusion detection systems

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ABSTRACT

The rapid increase in online risks is a reflection of the exponential growth of Internet of Things (IoT) networks. Researchers have proposed numerous intrusion detection techniques to mitigate the harm caused by these threats. Enterprises use intrusion detection systems (IDSs) and intrusion prevention systems (IPSs) to keep their networks safe, stable, and accessible. Network intrusion detection solutions have lately integrated powerful Machine Learning (ML) techniques to safeguard IoT networks. Selecting the proper data features for effectively training such ML models is critical to maximizing detection accuracy and computational efficiency. However, the efficiency of these systems degrades in high-dimensional data spaces, and it is crucial to have a suitable feature extraction method to eliminate extraneous data from the classification procedure. The detection accuracy and false positive rate of many ML-based IDSs also rise when the samples used to train the models are unbalanced. This study provides a detailed overview of the UNSW-NB15(DS-1) and NF-UNSWNB15(DS-2) datasets for intrusion detection, which will be utilized to develop and evaluate our models. In addition, this model uses the MaxAbsScaler algorithm to implement a filter-based feature scaling strategy. Then, use the condensed feature set to perform several ML techniques, including Support Vector Machines (SVM), K-nearest neighbors (KNN), Logistic Regression (LR), Naive Bayes (NB), Decision Tree (DT), and Random Forest (RF), considering multiclass classification. Accuracy tests for the multiclass classification scheme were improved from 60% to 94% using the MaxAbsScaler-based feature scaling method.

1. Introduction

The concept of the Internet of Things (IoT), referring to interconnected and varied networks of devices, is intimately associated with the digital revolution. The cybersecurity of modern network systems is seriously threatened by attacks from hostile third parties. The rapid advancement of communication networks, the Internet of Things (IoT), and other associated technologies has facilitated hackers in accelerating the rate of technological innovation in their tool sets. These fraudsters always seek novel methods to infiltrate secure computer systems. An Intrusion Detection System (IDS) is an essential component of any computer network. Intrusion Detection Systems (IDSs) are software or hardware systems designed to monitor a company's network for indications of potential or ongoing attacks. As per the 2023 Global Economic Forum Reports [1], Cybersecurity threats rank among the top five threats to the global economy at now. According to the study, organizations may soon encounter debilitating attacks that halt regular operations, leading to incomprehensible financial losses that surpass previous breaches. Ultimately, these debilitating intrusions would result in a significant rise in financing for the establishment of adequate

cyber security capabilities. The cyber security market has experienced a growth rate of over 30 times in the last 13 years, and it is expected to reach a spending level of \$133 billion by 2022 [2,3].

The Internet of Things (IoT) is a network of self-governing electronic and mechanical devices that have the ability to share data and communicate with each other and other devices through the internet [4]. Hackers are getting more cunning and hostile by the day. Treatpost, a leading IT news site, states that 98% of IoT device traffic is unencrypted and the majority of IoT gadgets (over 50%) are subject to an unacceptable threat [5,6]. In order to effectively combat the present environment of threats, which encompasses a wide range of scale, speed, and complexity, it is imperative that we possess the ability to respond with suitable countermeasures.

The seamless connectivity of IoT devices at elevated connections levels are upheld by their capacity to identify and prevent fraudulent traffic[7]. Preventing malicious traffic from accessing IoT devices is a crucial measure to safeguard them against attackers. Moreover, it is imperative to always protect the integrity of transmitted data[8]. The security of Internet of Things networks relies on promptly detecting DoS

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assaults. We consider all of these concerns and assess multiple machine learning algorithms using datasets from IoT-based Intrusion Detection Systems [9].

Many Intrusion Detection Systems (IDS) datasets are accessible for researchers to use in their studies. DARPA 98, KDD Cup 99, NSL-KDD, and UNSW-NB15 (DS-1) are the most popular [10–12]. The Cyber Range Lab at the Australian Centre for Cyber Security employs the IXIA PerfectStorm device to create the current real-scenario-based normal and attack traffic that comprises the DS-1. Furthermore, it encompasses nine prevalent assault techniques in addition to the anticipated traffic flow. Approximately 2.5 million intercepted packets are accessible to anybody. The dataset can be stored in formats such as BroIDS, csv, pcap, and argus [13,14]. From the entire dataset, the authors created .csv files that only include roughly 10% of the data, in addition to the full traffic. The most recent metric used to evaluate NIDS scenarios.

First, we looked into the various resources for investigating IDS and how it relates to the Internet of Things and found that the DS-2 dataset was the most suitable. The dataset is then pre-processed to make it more algorithm-friendly for execution.

We conducted experiments using kNN, DT, LR, NB, SVM and RF as supervised ML approaches for IDS. We evaluate the effectiveness of the chosen ML algorithms by employing methods over the NetFlow-based format of the DS-1, i.e., the DS-2.

The primary objectives of the proposed study are as follows:

1. To ascertain the nature of the data network traffic flow.
2. To Pursue Multi Class Classification of Network attacks.
3. To develop an effective IDS for the IoT network to limit information leakage on the test set utilizing the MaxAbsScaler-based feature scaling method.
4. To perform an experimental analysis against the dataset obtained from a traditional network setting that does not reflect contemporary attacks used in previous studies.

Following are the paper's remaining sections. The second section covers previous work. The many machine learning methods that contribute to this study are briefly discussed. The DS-1 and DS-2 is also described in depth. Additionally, during the experimental operations, the efficiency of several ML algorithms is determined. An overview of all the procedures and discussions involved in the experiments is then provided. The conclusion is the final part of this work.

2. Related work

With the goal of reducing the damage caused by cyberattacks, Ge et al. [15] established a methodology for tracing the origin of attacks. A sinkhole assault was a topic of this paper. However, the framework was not tested for its ability to detect attacks or generate false positives. A similar IDS for the IoT context was proposed by Mehare et al. [16], which takes into account the attacker nodes' locations and surrounding areas. Only denial-of-service attacks were discussed, and the proposed model was not tested.

Hodo et al. [17] deploy an ANN as a superior IDS for IoT to identify DoS/DDoS attacks by distinguishing regular behavior from attack patterns. In a virtual setting, the necessary dataset can be generated with a small number of instances. Using the UNSW-NB15 dataset, Koroniotis et al. [18] analyzed various machine learning approaches to determine their usefulness in monitoring harmful network activity and tracing botnets via unique IDs in network flows.

The NIDS presented by Samriya et al. [19] makes use of deep learning and algorithms inspired by nature. After the labels are converted to numbers, the dataset is normalized using the mini-max technique. It employs ant colony optimization to lower the data dimensionality and deep learning for training and testing the classification performance. The outcomes demonstrate a slight enhancement in comparison to the alternative models.

Angelin et al. [20] integrate convolutional neural network (CNN) and auto-encoder (AE) approaches to enhance the efficacy of the Network Intrusion Detection System (NIDS). The model's use of an autoencoder to reduce data characteristics enhances detection outcomes. Network traffic is only one example of the complicated data that convolutional neural networks can automatically extract patterns and characteristics from.

A Cost-based Random Forest Classifier (CRFC) approach has been introduced Pramilaranie et al. [21] to create a highly efficient intrusion detection system (IDS). By incorporating a cost matrix calculated from feature importance, the CRFC-based classification can be improved. This enhancement allows for better splitting of features, even in the presence of feature imbalance.

Samara et al. [22] discusses various intrusion detection techniques that have been developed for the IoT network. It is widely acknowledged that the Internet of Things (IoT) will have a significant impact on various aspects of our daily lives, encompassing education, vehicles, and healthcare. In addition, the wide range of connected devices and the need to maintain a high level of service pose a significant challenge.

Rehman et al. [23] examine various approaches to multiple intelligence methods and their implemented structures for network intrusion detection, with a specific emphasis on IoT threats and machine learning-based intrusion detection strategies. Baich et al. [24] carried out an analysis for both binary classification and multi-classification, with the latter involving five classes: Normal, DOS, Probe, U2R, and R2L.

Kumar et al. [25] conduct experiments on secure communication and measure the network's performance. By comparing these results with existing security metrics, propose a new and effective intrusion detection system (IDS). To enhance communication security, this IDS employs a deep learning-based classification approach known as fuzzy CNN.

Xu et al. [26] use the recursive feature elimination (RFE) algorithm and the binary grey wolf optimizer (BGWO) heuristic algorithm to pick the most important set of features for the target variable and reduce the number of features in the data. The synthetic minority oversampling technique (SMOTE) is employed to increase the representation of the minority class and alleviate the influence of data imbalances on the classification outcomes.

In order to demonstrate the increased complexity of the UNSW-NB15 dataset over the KDD99 dataset, Moustafa et al. [27] conducted a statistical study of the dataset and compared the results of several ML algorithms, concluding that the decision tree model delivers the best performance.

The KDD cup dataset favored IDS-based CNN over other IDS classifiers. It is important to minimize the number of false positives [28]. There has been a lot of work done on alarm filtering to increase detection accuracy. Intrusion detection and false alarm reduction were better with hybrid ant colony optimization than unsupervised clustering.

In order to achieve convergence, K-means clustering was applied to the ant colony optimization method [29]. Several ML models, including Recurrent Neural Networks and Feature Extraction with Independent Feature Selection (RNN-IDS), were investigated and their performance on the NSL-KDD dataset was analyzed. It is more vital to think about the time invested in training than on perfect accuracy [30]. SVM and extreme learning in multilayer-based hybrid IDS increases unknown attack detection. The redesigned K-means method improved IDS performance and training time. [31].

Mbona et al. [32], the authors investigate a method based on machine learning. Experiments showed that ML techniques, when used in conjunction with flow IDs, were successful in identifying botnet intrusions. We employed ARM, ANN, Naive Bayes, and Decision Tree, four machine learning techniques. Based on the findings, the Decision Tree performed best with a 93.23% accuracy and a 6.77% False Alarm Rate, while the ANN performed worst with a 63.97% accuracy and a 36.03% False Alarm Rate.

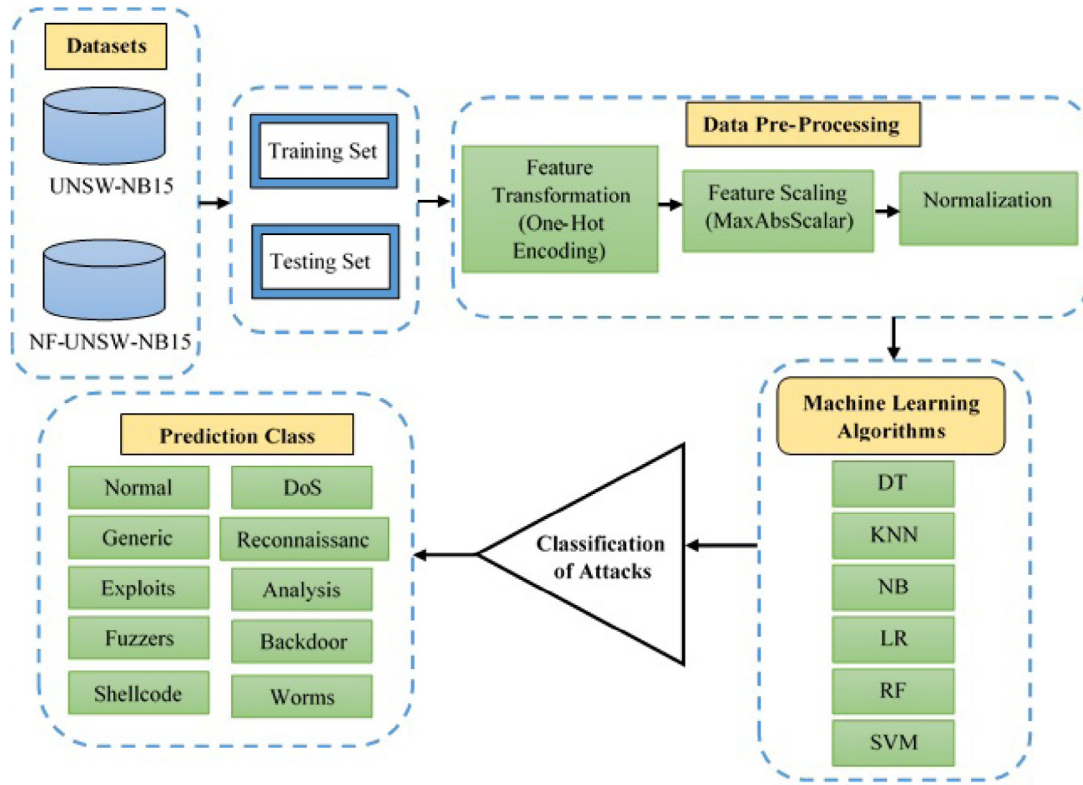


Fig. 1. ML Based IDS architecture.

3. Proposed system and materials

This specific effort will employ a method that entails the following distinct phases to identify the network intrusion: (i) Cleaning and preparing the data, (ii) choosing the features, and (iii) classifying the data using CM models. An intricate diagram showing the whole process flow of the suggested methodology can be found in Fig. 1.

3.1. Data pre-processing

Both pre-processing and data cleaning mean the same thing. To substitute unnecessary or low-IG attributes, features generated from other features in the data are used. The elimination or replacement of null values is necessary for certain ML models that require particular information, like RF methods. Imputation provides a workable solution to this problem. Adding insult to injury, certain ML techniques can only process numeric data types, like integers and floats. This issue can be resolved by either typecasting the values or disabling the inconsistent functionality. In order to guarantee consistency and decrease computing complexity, it is essential that data be compatible with many methods during pre-processing.

3.2. Feature normalization

Numerous features, such as SVM, LR, DT, RF, NB, and kNN, have a significant numerical value that impacts the learning process in machine learning. It also takes a lot of computing power to train on high-dimensional datasets. As a result, data is often scaled using methods such as Z-score standardization, Decimal scaling, Max normalization, Min–Max scaling, and Max absolute scaling in order to mitigate these issues. The task at hand should be considered before selecting a method. Here during the data analysis stage, we use Max Absolute Scaling Eq. (1).

Each feature's maximum value in the training set should be set to 1.0, this estimator scales and translates each feature independently.

Our targets' labeled features are encoded using one-hot encoding for supervised learning, which does not shift or center the data, preserving sparsity.

$$X_{scaling} = \frac{X}{Max(X)} \quad (1)$$

3.3. Datasets

3.3.1. UNSW-NB15 (DS-1) dataset

The Australian Cyber Security Center's Cyber Lab developed the DS-1 dataset. The primary goal of the collection is to obtain realistic examples of everyday behavior alongside artificial examples of sophisticated attack techniques [33]. There are over 2 million records in the dataset, and 49 distinct features were retrieved using advanced techniques. The training set has 175,341 records, while the testing set contains 82,332 records of two types: attack and normal. There is a clear dichotomy between typical and unusual data in this group. However, it has nine distinct forms of assault [34] (see Table 1).

3.3.2. NF-UNSW-NB15 (DS-2) dataset

The DS-1 NetFlow representation, DS-2, has been created and annotated with attack types. 1,623,118 data flows were counted in total, with 1,550,712 (95.54%) being safe and 72,406 (4.46%) being attack samples. Distribution of all flows in the DS-2 and DS-1 are shown in Table 2. The attack samples are further divided into nine groups and the correlation matrix of both datasets are depicted in Fig. 2.

4. Methodology

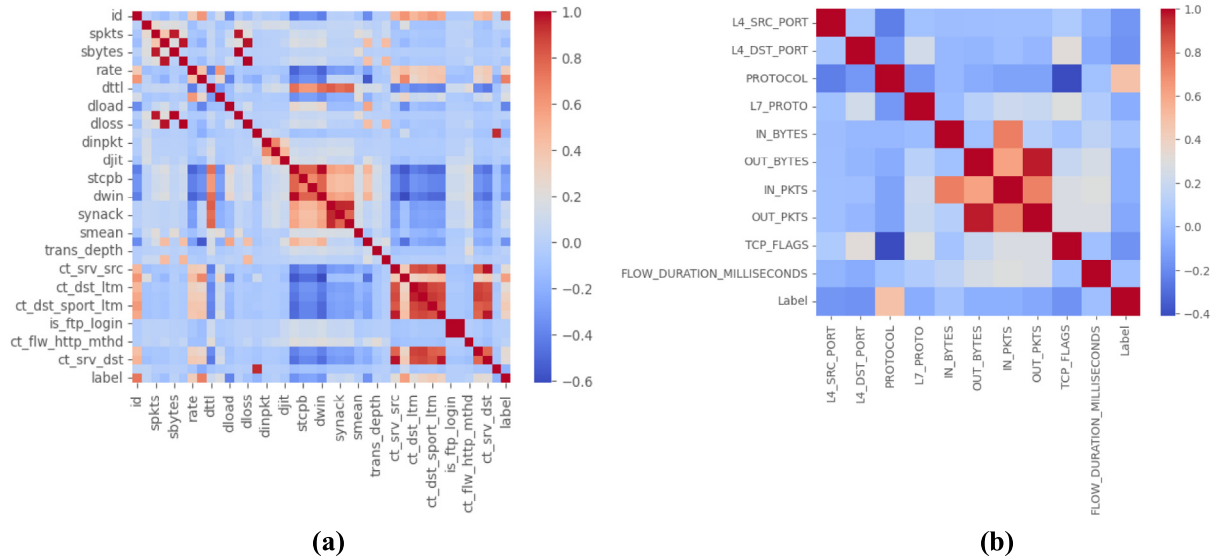
The main objective of any classification challenge is to find the optimal class label for each instance in the test dataset. Classification approaches are used to build models that can correctly identify classes in a given dataset. The trials are carried out using ML techniques such as DT, RF, NB, LR, KNN, and SVM. Adopting these tactics is a popular way to produce efficient and effective IDS. Here we will review all of

Table 1
Dataset description.

Type	Description
Normal	Observations of actual transaction.
Generic	It does not matter how a block-cipher is constructed; the technique will work against all block-ciphers (of a certain block and key size).
Exploits	Exploit code is used by attackers to take advantage of vulnerabilities in a target operating system or application.
Fuzzers	Flooding a system with random data to crash it.
DoS	An attempt to prevent users from accessing a server or other network resource, generally by disabling or interrupting those services on an Internet-connected host.
Reconnaissance	Includes all Strikes that can be used to imitate information-gathering attacks.
Analysis	Some of the attacks described include port scanning, spamming, and breaking into .html files.
Backdoor	The process of stealthily accessing a computer or its data in order to steal or modify it.
Shellcode	The main piece of malicious code utilized in exploiting a software flaw.
Worms	In order to infect other machines, the attacker replicates itself. It can easily spread from one machine to another, entering its intended host machine through a security hole.

Table 2
Distribution of training and testing datasets.

Type of attacks	DS-1		DS-2	
	Training data	Testing data	Training data	Testing data
Normal	56,000	37,000	42558	10 631
Generic	40,000	18,871	381	73
Exploits	33,393	11,132	1254	330
Fuzzers	18,184	6,062	664	165
DoS	12,264	4,089	358	105
Reconnaissance	10,491	3,496	690	163
Analysis	2,000	677	196	57
Backdoor	1,746	583	216	56
Shellcode	1,133	378	61	14
Worms	130	44	8	3
Total Samples	1,75,341	82,332	46,386	11,597

**Fig. 2.** Correlation Matrix of (a) DS-1 dataset (b) DS-2 dataset.

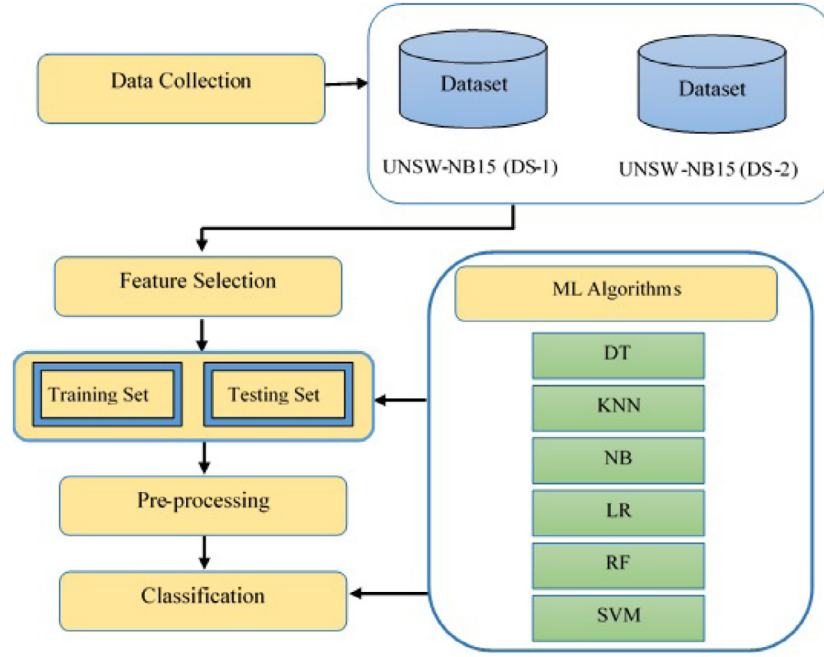


Fig. 3. Overview of the Study Methodological Framework.

the machine learning methods used to categorize attacks. Meanwhile, the testing set functions as the arena for assessing the performance of the classifier. Metrics such as accuracy, precision, recall, and F1-score are used to evaluate their effectiveness, resulting in the choice of a particularly effective classifier for further study, as shown in Fig. 3.

4.1. Naïve Bayes(NB)

Assuming that the predictors are independent, Bayes' Theorem offers a way to categorize. In its most basic form, a Naive Bayes classifier operates on the premise that the existence or lack of a feature within a class does not influence the existence or absence of any other feature. Text classification is the main application of NB. Classification and grouping are two of the most common uses of the conditional probability of an event. Bayes' Theorem is given in Eqs. (2) and (3)

Bayes' Theorem:

$$P(C_p | P_r) = \frac{[P(P_r | C_p) * P(C_p)]}{P(P_r)} \quad (2)$$

$$P(C_p | P_r) = P(C_{p1} | P_r) \times P(C_{p2} | P_r) \times \dots \times P(C_{pn} | P_r) \times P(P_r) \quad (3)$$

Where :-

$P(C_p | P_r)$: Posterior Probability

$P(P_r | C_p)$: Likelihood

$P(X)$: Class Prior Probability

$P(P_r)$: Predictor Prior Probability

4.2. SVM

As a supervised learning strategy, SVM is best used for assessing data in the context of classification and regression. Non-linear classification is possible using support vector machines thanks to the kernel approach for mapping inputs onto high-dimensional feature spaces [35]. It is a line that separates people of different economic backgrounds. Misclassification is reduced by keeping the edges as far from the classes as practicable [36,37].

SVM is typically employed for binary classification, where information is simply sorted into 1 or 0 categories. Multiclass classification makes use of the same principle. The multiclass problem can be broken

down into a series of binary classification issues, sometimes known as one-on-one cases [38]. One-on-one must be specified explicitly, as can be seen further down the code, as it is not the default in scikit-learn. A one-on-one setting is the default. Class x data is isolated from the rest of the data. One group is repeatedly highlighted as being different from the rest. Scikit-learn has a wide variety of popular models, including linear, polynomial, radial basis function, and sigmoid.

Since it closely resembles the Gaussian distribution, RBF kernels are frequently employed. In addition, they are the most general kernelization type. The RBF kernel function calculates a distance between two points P_1 and P_2 given their coordinates. This kernel's mathematical expression is given in Eq. (4):

$$K(P_1, P_2) = e^{-(\gamma \|P_1 - P_2\|^2)} \quad (4)$$

Here, $\|P_1 - P_2\|$ is the Euclidean (L2-norm) Distance between the two points P_1 and P_2 , and ' γ ' chooses which training samples have an impact on the decision boundary.

4.3. Random forest (RF)

RF is an ML supervised learning algorithm for classification and regression. This method uses the combined results of numerous classifiers to get more reliable predictions [39]. To improve the model's predictive performance, we incorporate many decision trees for different parts of the input information and take an average. RF is commonly employed in ML due to its speed and ability to handle missing and imperfect input. It is possible that certain decision trees will produce the correct answers while others will not when using the RF to forecast the class of the dataset [40]. The trees only provide reliable forecasts, however, when taken as a whole. When working with Random Forest, the importance of a feature f_i is determined by averaging its weight over all trees. The total feature relevance is calculated using Eq(5) across all trees, and then the number of trees is divided by that amount:

$$RFI(f_i) = \frac{1}{T_i} \sum_t \sum_k \frac{T_i}{T} \Delta I_{f_i} \quad (5)$$

Here, T_i stands for the total number of trees, the number of samples that reach node k is denoted as N_k , t is the total no of samples present

in the dataset and I represents the gini index value which is calculated by Eq (6).

$$GiniIndex(I) = 1 - \sum_{i=1}^n (P_i)^2 = 1 - [(P_p)^2 + (P_n)^2] \quad (6)$$

Here, P_p represents the positive class and P_n represents the negative class.

4.4. Logistic regression (LR)

Although it is called a “regression” technique, LR is most commonly used for the binary classification problem. Multiclass classification problems that use one-vs-rest learning algorithms can also benefit from the LR’s use [41,42]. LR models use the sigmoid function or variants on linear ML models. The result of this process is a compressed range between zero and one. It is more likely that a particular class exists if the resulting value is closer to 1. Eq. (7) gives the mathematical formulation.

$$LogisticRegression : \sigma(x) = \frac{1}{1 + e^{-x}} \quad (7)$$

In the above expression ‘ σ ’ operates on x .

4.5. Decision tree (DT)

One supervised ML method used for both classification and regression is the decision tree. The model that emerged from the deliberation took the form of a tree. Because of this, DT is easier to grasp for end users. In addition, several ML programs make it possible to view the generated trees graphically [43,44]. A measure of the likelihood of an attack class being misclassified when randomly selected is entropy, which is represented in Eq.(8)

$$E = \sum_{n=1}^c -p_n \log(p_n) \quad (8)$$

where c is the number of classes and p_m is the probability of data being categorized to a specific class of n .

4.6. K-nearest neighbors(KNN)

Both supervised and unsupervised tasks are within the capabilities of the k-nearest neighbors ML algorithm. Many popular clustering methods nowadays, for instance, have their roots in KNN [45]. The KNN supervised ML variant was used in this study. This method relies on the commonly used Euclidean (EM) metric as its foundation. The EM measures the separation between two locations (instances) in three-dimensional space [46]. The EM between two points p and q in a Euclidean space Z is calculated as given in Eq. (9).

$$\Delta(p, q) = \sqrt{\sum_{i=1}^r (p_k - q_k)^2} \quad (9)$$

The value r indicates how many times Z can occur at most. In order to determine the identity (label) of an instance r_0 in Z , the kNN method calculates the EM separating r_0 from its k-Nearest-Neighbors examples in Z , and then chooses the label (class) of r_0 based on the class of its k related neighbor instances.

4.7. Evaluation metrics

The confusion matrix helps evaluate model performance. It details the proportion of accurate forecasts, as well as the number of erroneous ones (both FSP and FSN). The model’s efficacy is evaluated in a number of ways, including:

- i Time: Algorithms are measured on how quickly they can run a given machine learning model. We have already established that a slow-running algorithm may not thrive on the IoT.

Table 3

Example of the confusion matrix.

Predicted/Actual	Positive(P) (0)	Negative(N) (1)
Positive(P)(0)	TRP	FSP
Negative(N)(1)	FSN	TRN

Table 4

Classification report of DT model for DS-1.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	423
1	0.01	0.01	0.01	362
2	0.10	0.19	0.13	2459
3	0.53	0.37	0.44	6617
4	0.28	0.26	0.27	3598
5	0.01	0.00	0.00	7988
6	0.73	0.91	0.81	11199
7	0.02	0.08	0.04	2174
8	0.14	0.19	0.16	231
9	0.25	0.06	0.09	18
Acc.			0.41	35069
macro avg	0.21	0.21	0.19	35069
Wegt. avg	0.37	0.41	0.38	35069

Table 5

Classification report of kNN model for DS-1.

	Pr	Rc	F1-Sc	Supp.
0	0.16	0.20	0.18	423
1	0.04	0.01	0.02	362
2	0.26	0.31	0.28	2459
3	0.58	0.60	0.59	6617
4	0.45	0.41	0.43	3598
5	1.00	0.98	0.99	7988
6	0.81	0.89	0.85	11199
7	0.47	0.25	0.33	2174
8	0.09	0.03	0.05	231
9	0.00	0.00	0.00	18
Acc.			0.70	35069
macro avg	0.39	0.37	0.37	35069
Wegt. avg	0.69	0.70	0.69	35069

- ii Accurate Positive Predictions (True Positives): This is to be expected when the model accurately predicts a positive class. Inaccurate predictions of positive classes by models are called false positives.
- iii Confusion Matrix: “Confusion matrices” are used to summarize the model’s predictions for each category. This facilitates the preparation of a model presentation, as shown in Table 3. This matrix has the size NN , where N is the total number of classes for a given set of components, with N rows representing expected classes and N columns representing observed classes. Figs. 4 and 5 shows the confusion matrix of different machine learning models for UNSW-NB15 (DS-1) and NF-UNSW-NB15 (DS-2) datasets respectively.
- iv Precision: There is just a simple count of “how many relevant data items” displayed. How much of an algorithm’s supposed positive observations are correct? The proportion of true positives to all true and false positives is a good indicator of accuracy.

$$Precision(Pr) = \frac{TRP}{TRP + FSP} \quad (10)$$

- v Recall: An estimate of the number of correctly detected events or the true positive rate.

$$Recall(Rc) = \frac{TRP}{TRP + FSN} \quad (11)$$

Table 6
Classification report of LR model for DS-1.

	Pr	Rc	F1-Sc	Supp.
0	0.11	0.25	0.15	423
1	0.00	0.00	0.00	362
2	0.02	0.01	0.01	2459
3	0.36	0.27	0.31	6617
4	0.46	0.46	0.46	3598
5	0.95	0.95	0.95	7988
6	0.91	0.51	0.66	11 199
7	0.18	0.84	0.29	2174
8	0.00	0.00	0.00	231
9	0.00	0.00	0.00	18
Acc.			0.54	35 069
macro avg	0.30	0.33	0.28	35 069
Wegt. avg	0.64	0.54	0.55	35 069

Table 7
Classification report of NB model for DS-1.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	423
1	0.00	0.00	0.00	362
2	0.43	0.00	0.01	2459
3	0.45	0.36	0.40	6617
4	0.00	0.00	0.00	3598
5	0.00	0.00	0.00	7988
6	0.38	1.00	0.55	11 199
7	0.00	0.00	0.00	2174
8	0.00	0.00	0.00	231
9	0.00	0.00	0.00	18
Acc.			0.39	35 069
macro avg	0.13	0.14	0.10	35 069
Wegt. avg	0.23	0.39	0.25	35 069

Table 8
Classification report of RF model for DS-1.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	423
1	0.27	0.01	0.02	362
2	0.21	0.10	0.14	2459
3	0.46	0.24	0.31	6617
4	0.55	0.10	0.18	3598
5	0.99	0.98	0.98	7988
6	0.51	0.98	0.67	11 199
7	0.44	0.04	0.08	2174
8	0.16	0.14	0.15	231
9	0.00	0.00	0.00	18
Acc.			0.60	35 069
macro avg	0.36	0.26	0.25	35 069
Wegt. avg	0.58	0.60	0.53	35 069

Table 9
Classification report of SVM model for DS-1.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	423
1	0.00	0.00	0.00	362
2	0.00	0.00	0.00	2459
3	0.19	1.00	0.32	6617
4	0.00	0.00	0.00	3598
5	0.00	0.00	0.00	7988
6	0.49	0.00	0.01	11 199
7	0.00	0.00	0.00	2174
8	0.00	0.00	0.00	231
9	0.00	0.00	0.00	18
Acc.			0.19	35 069
macro avg	0.07	0.10	0.03	35 069
Wegt. avg	0.19	0.19	0.06	35 069

Table 10
Classification report of DT model for DS-2.

	Pr	Rc	F1-Sc	Supp.
0	0.13	0.05	0.07	57
1	0.09	0.86	0.17	56
2	0.99	0.98	0.98	10 631
3	0.06	0.17	0.08	105
4	0.72	0.10	0.18	330
5	0.19	0.03	0.05	165
6	0.08	0.14	0.10	73
7	0.00	0.00	0.00	163
8	0.12	0.14	0.13	14
9	0.00	0.00	0.00	3
Acc.			0.91	11 597
macro avg	0.29	0.31	0.18	11 597
Wegt. avg	0.94	0.93	0.91	11 597

Table 11
Classification report of kNN model for DS-2.

	Pr	Rc	F1-Sc	Supp.
0	0.18	0.11	0.13	57
1	0.14	0.80	0.23	56
2	0.99	0.98	0.98	10 631
3	0.11	0.01	0.02	105
4	0.46	0.60	0.53	330
5	0.45	0.28	0.35	165
6	0.08	0.05	0.06	73
7	0.48	0.26	0.33	163
8	0.00	0.00	0.00	14
9	0.00	0.00	0.00	3
Acc.			0.93	11 597
macro avg	0.29	0.31	0.26	11 597
Wegt. avg	0.94	0.93	0.93	11 597

- vi F1-Score: The F1-score is better since it calculates the harmonic mean of precision and recall using both false positives and negatives.

$$F1 - Score(F1_{sc}) = 2 * \frac{Pr * Rc}{Pr + Rc} \quad (12)$$

- vii Accuracy: A measure of how many instances in a test dataset were correctly labeled relative to the total number of instances in the dataset. Here's how it is graded:

$$Accuracy(Acc) = \frac{TRP + TRN}{TRP + FSP + TRN + FSN} \quad (13)$$

5. Result analysis and discussions

5.1. UNSW-NB15 (DS-1) dataset

Tables 4 to 9 show the outcomes of the multiclass classification methodology that utilized the whole feature space of the DS-1, as determined by the ML methods.

5.2. NF-UNSW-nb15(DS-2) dataset

Tables 10 to 15 show the outcomes of the multiclass classification methodology that utilized the whole feature space of the DS-2, as determined by the ML methods.

5.3. Simulations:

The experiments for this study were conducted using Python 3.8 on a Windows 11 machine equipped with a Core i5 CPU and 16 GB of RAM.

Typical receiver performance measured by the receiver's (AUC-ROC) curve. This curve measures the effectiveness of a classification problem at different thresholds. ROC is a probability curve, while AUC is a metric for assessing how well two groups may be distinguished. The model's ability to differentiate between different types of attacks is indicated by this metric. The AUC assesses a model's ability to differentiate between 0 and 1 classes. The AUC measures how well a model can tell the difference between an attack and a normal situation.

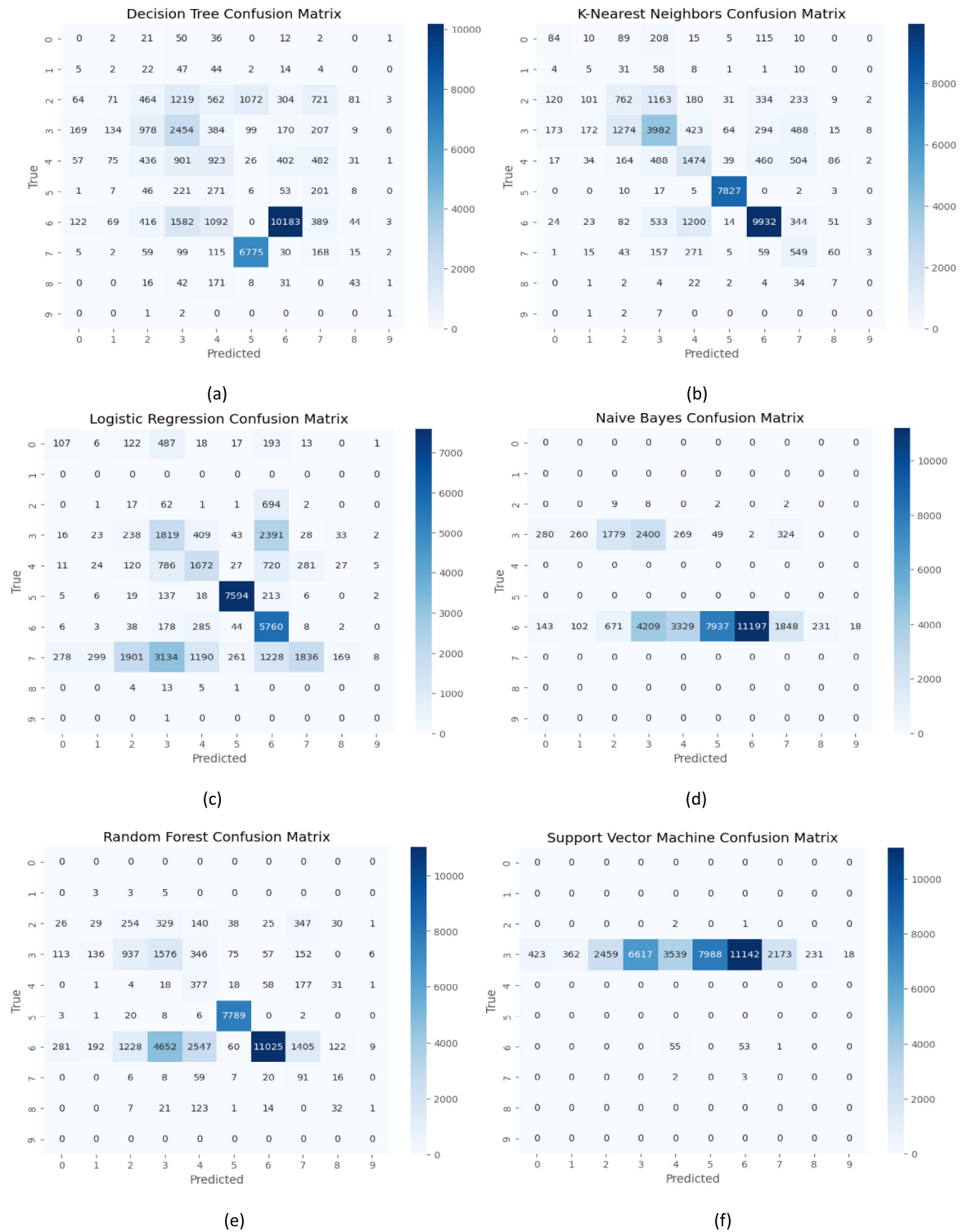


Fig. 4. Confusion Matrix of different ML models for UNSW-NB15 (DS-1) Dataset.

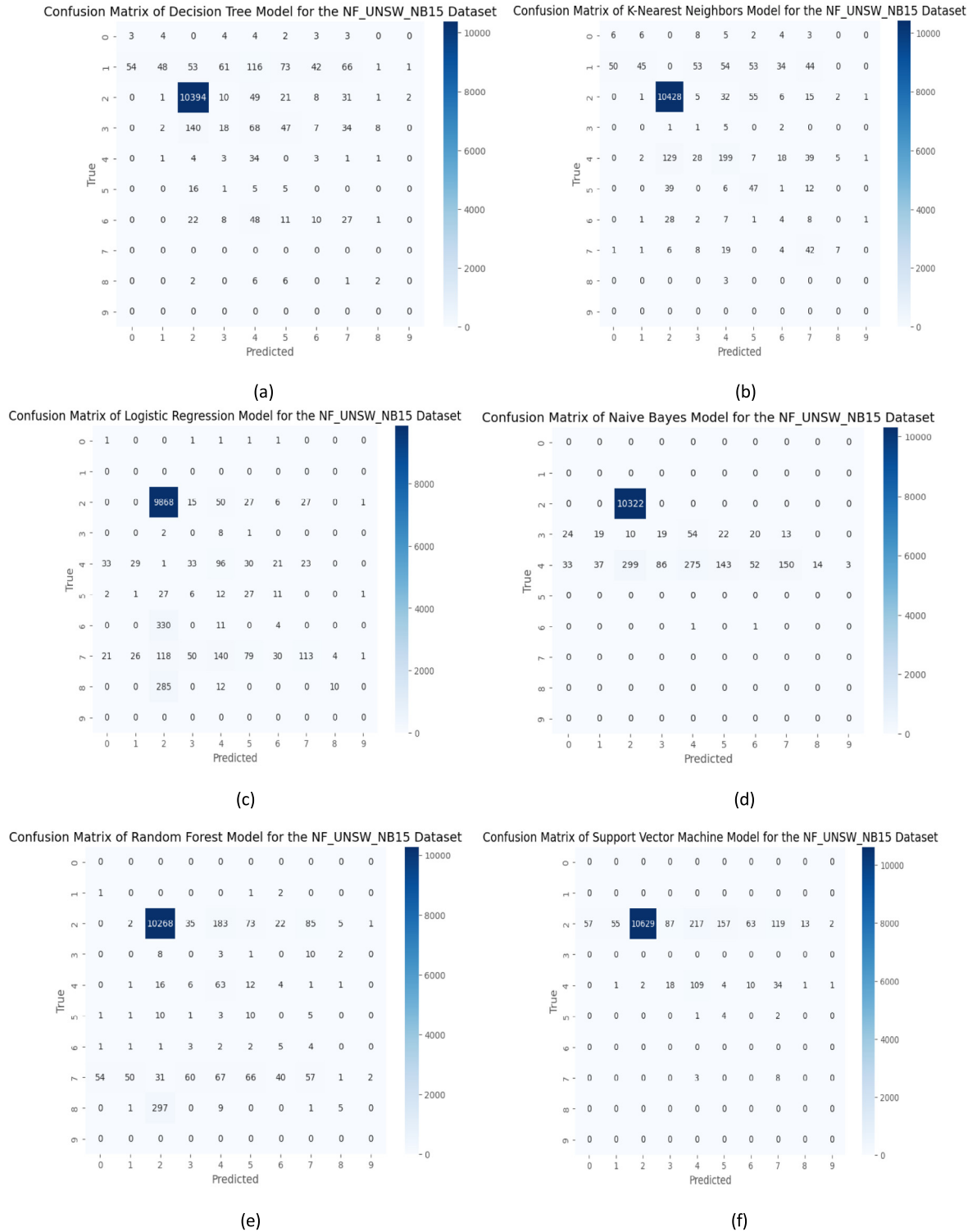


Fig. 5. Confusion Matrix of different ML models for NF-UNSW-NB15 (DS-2) Dataset.

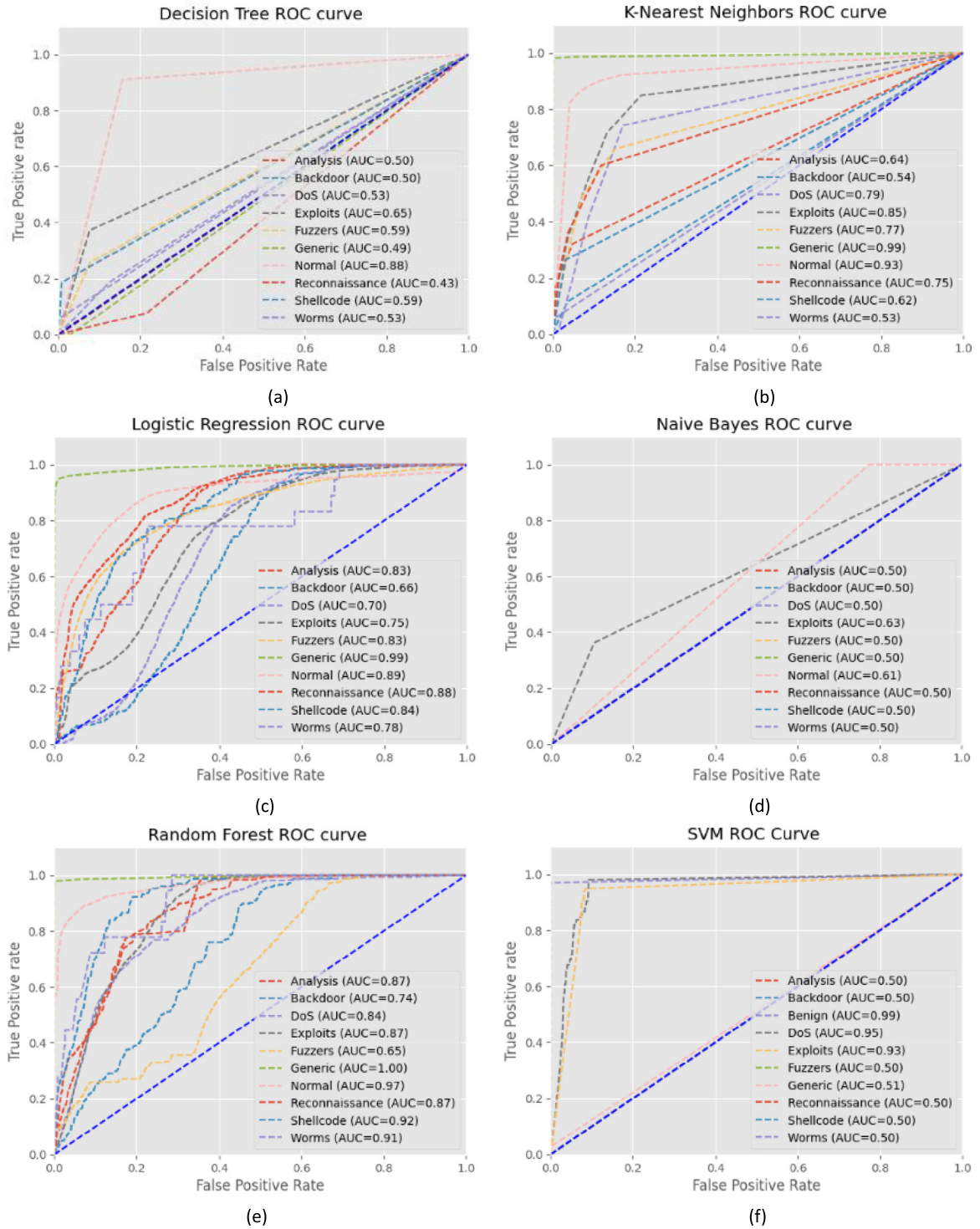
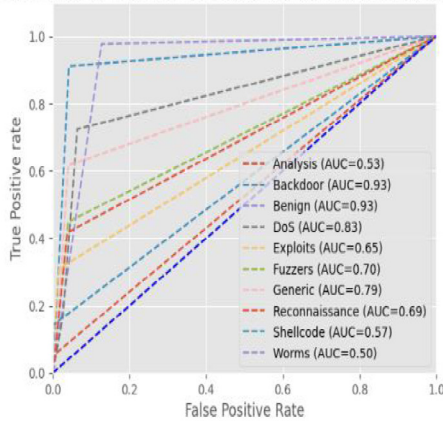


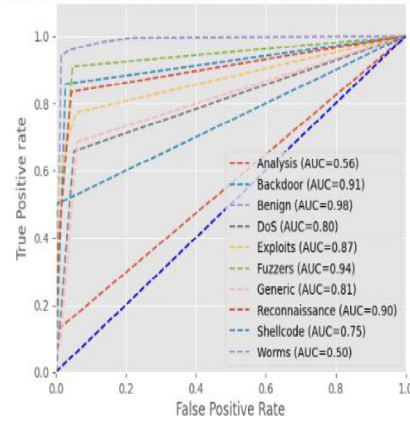
Fig. 6. ROC and AUC Curve of different ML models for UNSW-NB15 Dataset.

ROC-AUC curve of Decision Tree Model for the NF_UNSW_NB15 Dataset



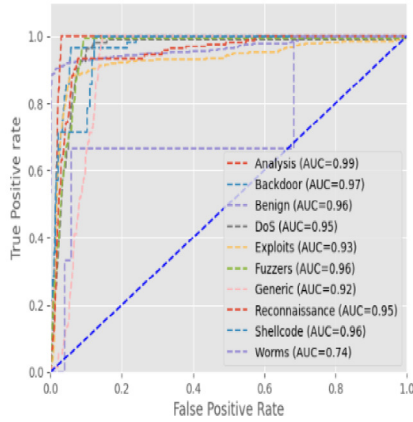
(a)

ROC-AUC curve of K-Nearest Neighbors Model for the NF_UNSW_NB15 Dataset



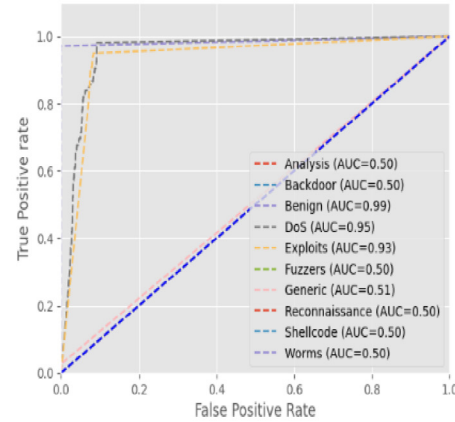
(b)

ROC-AUC curve of Logistic Regression Model for the NF_UNSW_NB15 Dataset



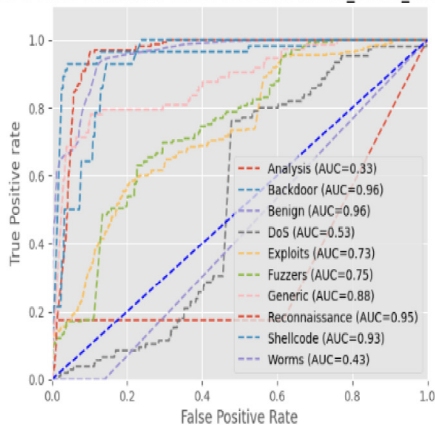
(c)

ROC-AUC curve of Naive Bayes Model for the NF_UNSW_NB15 Dataset



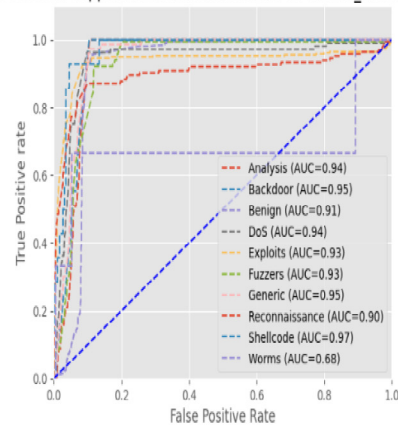
(d)

ROC-AUC curve of Random Forest Model for the NF_UNSW_NB15 Dataset



(e)

ROC-AUC curve of Support Vector Machine model for the NF_UNSW_NB15 Dataset



(f)

Fig. 7. ROC and AUC Curve of different ML models for NF-UNSW-NB15 (DS-2) Dataset.

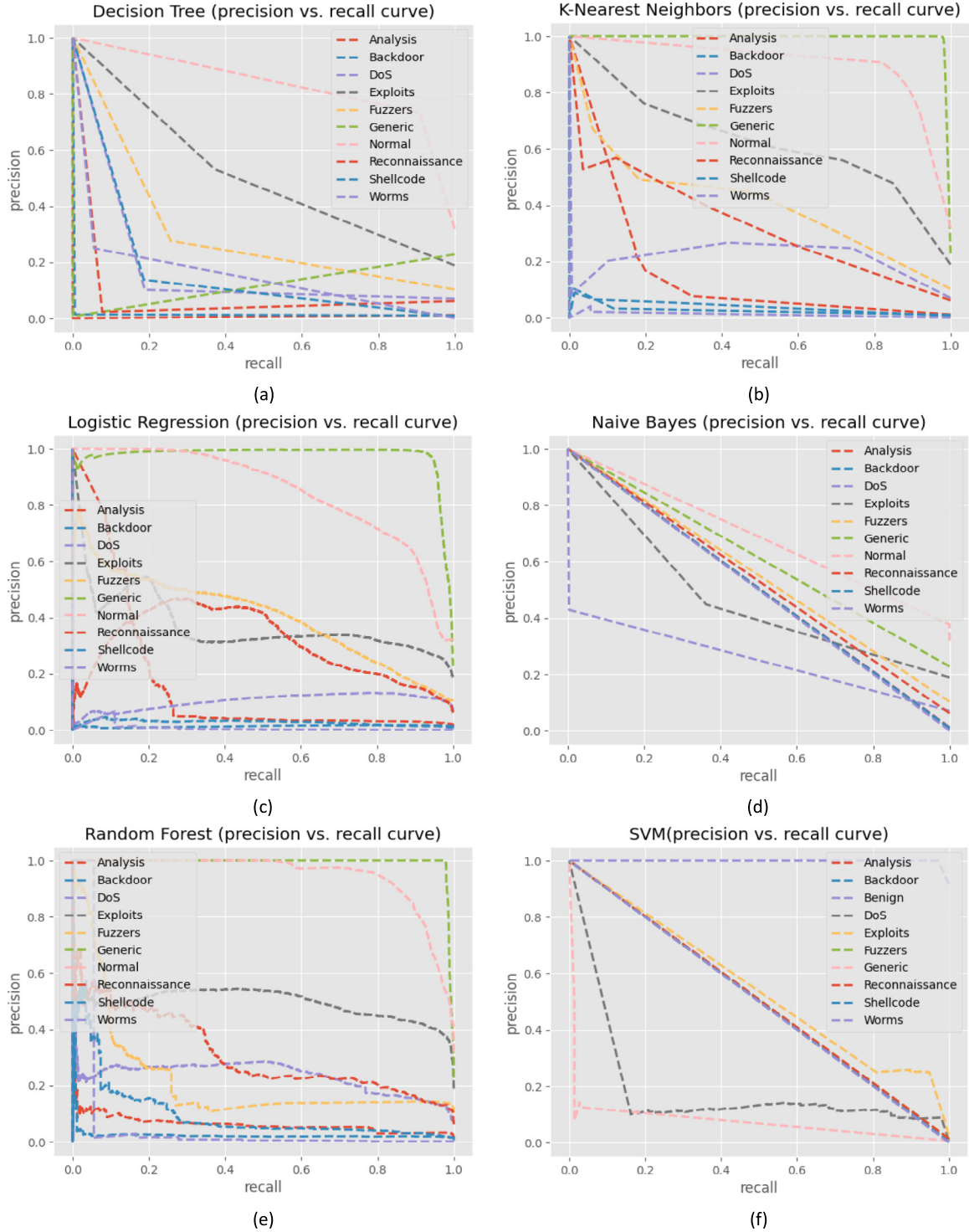


Fig. 8. Precision and Recall Curve of different ML models for UNSW-NB15 (DS-1) Dataset.

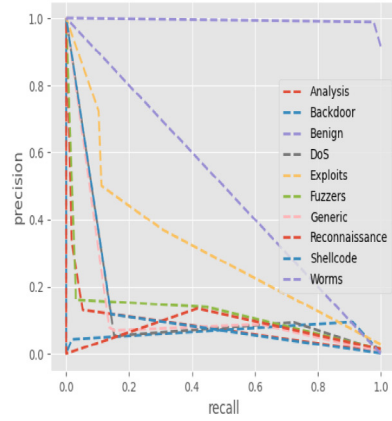
TPR is plotted against FPR on the ROC curve, which places TPR on the y-axis and FPR on the x-axis. Fig. 6 shows the results of the machine learning models on UNSW-NB15 dataset and Fig. 7 shows on NF-UNSW-NB15 dataset in terms of ROC-AUC curves.

When there is a large disparity between the classes, the Precision-Recall metric is a good indicator of how well the prediction was made. Information retrieval metrics include accuracy (how accurate the results are) and recall (how many accurate results are returned). The precision-recall curve illustrates the compromise between the two metrics at certain thresholds. In this context, a low false positive

rate corresponds to a high area under the curve, while a high recall rate corresponds to a low false negative rate. Accurate results (high accuracy) and a large proportion of positive outcomes (high recall) indicate that the classifier is doing well.

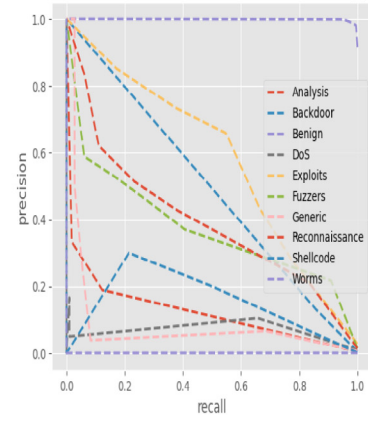
When comparing the system's predicted labels to the training labels, if the recall is high but the precision is low, the system will return many false positives. One can expect fewer results from a system with high precision and low recall but, when compared to the training labels, its predictions are more likely to be accurate. In a perfect system, numerous results would be returned, and the labels would be right.

Precision vs. Recall curve of Decision Tree Model for the NF_UNSW_NB15 Dataset



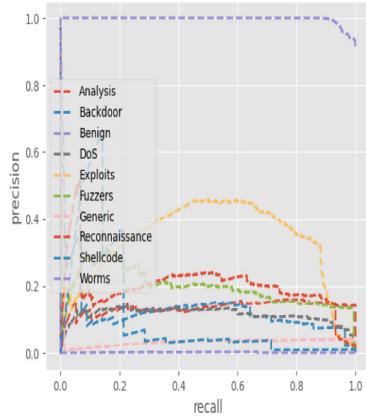
(a)

Precision vs. Recall curve of K-Nearest Neighbors Model for the NF_UNSW_NB15 Dataset



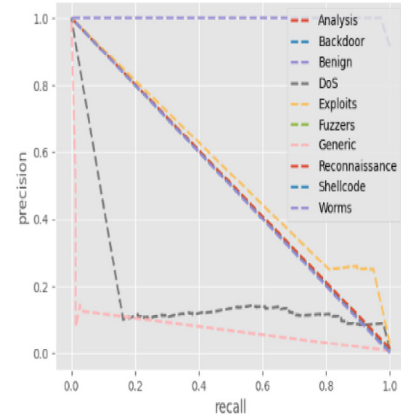
(b)

Precision vs. Recall curve of Logistic Regression Model for the NF_UNSW_NB15 Dataset



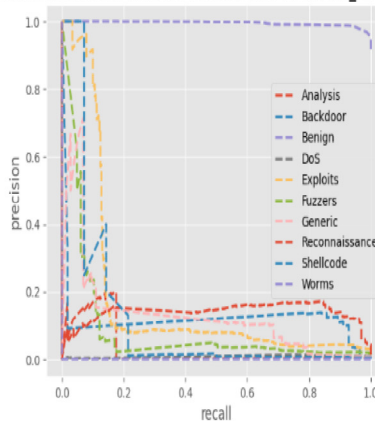
(c)

Precision vs. Recall curve of Naive Bayes Model for the NF_UNSW_NB15 Dataset



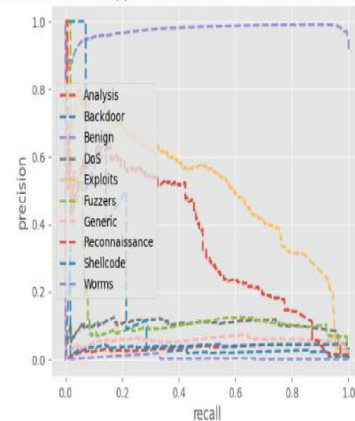
(d)

Precision vs. Recall curve of Random Forest Model for the NF_UNSW_NB15 Dataset



(e)

Precision vs. Recall curve of Support Vector Machine model for the NF_UNSW_NB15 Dataset



(f)

Fig. 9. Precision and Recall Curve of different ML models for NF-UNSW-NB15 (DS-2) Dataset.

Table 12
Classification report of LR model for DS-2.

	Pr	Rc	F1-Sc	Supp.
0	0.18	0.11	0.13	57
1	0.14	0.80	0.23	56
2	0.99	0.98	0.98	10 631
3	0.11	0.01	0.02	105
4	0.46	0.60	0.53	330
5	0.45	0.28	0.35	165
6	0.08	0.05	0.06	73
7	0.48	0.26	0.33	163
8	0.00	0.00	0.00	14
9	0.00	0.00	0.00	3
Acc.			0.93	11 597
macro avg	0.29	0.31	0.26	11 597
Wegt. avg	0.94	0.93	0.93	11 597

Table 13
Classification report of NB model for DS-2.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	57
1	0.00	0.00	0.00	56
2	1.00	0.97	0.99	10 631
3	0.10	0.18	0.13	105
4	0.25	0.83	0.39	330
5	0.00	0.00	0.00	165
6	0.50	0.01	0.03	73
7	0.00	0.00	0.00	163
8	0.00	0.00	0.00	14
9	0.00	0.00	0.00	3
Acc.			0.92	11 597
macro avg	0.19	0.20	0.15	11 597
Wegt. avg	0.93	0.92	0.92	11 597

Table 14
Classification report of RF model for DS-2.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	57
1	0.00	0.00	0.00	56
2	0.96	0.97	0.96	10 631
3	0.00	0.00	0.00	105
4	0.61	0.19	0.29	330
5	0.32	0.06	0.10	165
6	0.26	0.07	0.11	73
7	0.13	0.35	0.19	163
8	0.02	0.36	0.03	14
9	0.00	0.00	0.00	3
Acc.			0.90	11 597
macro avg	0.23	0.20	0.17	11 597
Wegt. avg	0.91	0.90	0.90	11 597

Table 15
Classification report of SVM model for DS-2.

	Pr	Rc	F1-Sc	Supp.
0	0.00	0.00	0.00	57
1	0.00	0.00	0.00	56
2	0.93	1.00	0.96	10 631
3	0.00	0.00	0.00	105
4	0.61	0.33	0.43	330
5	0.57	0.02	0.05	165
6	0.00	0.00	0.00	73
7	0.73	0.05	0.09	163
8	0.00	0.00	0.00	14
9	0.00	0.00	0.00	3
Acc.			0.93	11 597
macro avg	0.28	0.14	0.15	11 597
Wegt. avg	0.89	0.93	0.90	11 597

Table 16
Result analysis of UNSW-NB15 (DS-1) dataset.

Methods	Time cost (s)	Testing acc.
Decision Tree (DT)	4.6190	0.41
K-Nearest Neighbor (KNN)	3060.3462	0.70
Logistic Regression	23.9960	0.54
Naïve Bayes (NB)	6.1234	0.39
Random Forest (RF)	95.7328	0.60
Support Vector Machine (SVM)	8849.0350	0.19

Table 17
Result analysis of NF-UNSW-NB15 (DS-2) dataset.

Methods	Time cost (s)	Testing accuracy
Decision Tree (DT)	0.2137	0.91
K-Nearest Neighbor (KNN)	50.7952	0.93
Logistic Regression	5.2340	0.93
Naïve Bayes (NB)	0.1495	0.92
Random Forest (RF)	18.1105	0.94
Support Vector Machine (SVM)	529.6916	0.93

Fig. 8 shows the results of the machine learning models on UNSW-NB15 (DS-1) dataset and Fig. 9 shows on NF-UNSW-NB15 (DS-2) dataset in terms of Precision and Recall curves.

6. Conclusion and future work

In order to create trustworthy IDSs, this research looked at the possibility of using various ML techniques in tandem with the MaxAbsScalar algorithm for feature scaling. The DS-1 was used to assess these methods. In this study, we looked at the multiclass categorization scenarios. In addition, 19 very beneficial characteristics were chosen from both datasets using an attribute selection technique based on MaxAbsScalar. We looked at how other studies have used the DS-1 with different feature selection approaches during our exhaustive literature analysis, which helped put our findings into context. In addition to summarizing the results, we compared them to those of other classifiers in the literature and our proposed technique.

At first, we conducted the trials across the entire feature space of both the datasets using the recommended ML techniques. The experiments were then conducted utilizing the MaxAbsScalar feature scaling algorithm's output reduced feature vector. The data from the experiments indicated that the complexity of the models could be lowered and the detection accuracy on test data could be improved by utilizing a reduced (optimal) feature vector. KNN, LR, NB, DT, SVM, and RF ML classifiers were tested for intrusion detection in the proposed taxonomy.

Both the data-sets are used to evaluate the classifiers. Precision, recall, F1-Score, accuracy, TPR, and FPR are used to evaluate the classifiers. With the right settings, the RF classifier outperforms the competition on the UNSW-dataset. The results in Table 16, Table 17 show that the RF classifier has an accuracy of 60% and 94% for DS-1 and DS-2 respectively. In Table 18 we conducted a comparative analysis with other established methods employed for data preparation. This study can be expanded in the future to include intrusion detection using feature analysis mechanisms and selected characteristics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Table 18

Comparison of the proposed model with existing approaches (UNSW_NB15 and NF-UNSW_NB15 Dataset).

Reference	Data preprocessing technique	Classification algorithm	Accuracy
[47]	Categorical features remove redundant and irrelevant features	Classification and Regression Tree (CART)	87.74%
[48]	Data normalization and Data transformation	Embedding SVM Naive Bayes	93.75%
[49]	Object features encoding and Data Normalization	Two phase IDS using Naive Bayes	86.90%
Proposed approach	Feature Transformation, Feature Scaling and Data Normalization	DT,KNN, LR, NB, RF, SVM	94.00%

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